

SIMULATION REPORT

Simulation Results and Analysis

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Experimental Conditions

Common setup

- Simulator: LTspice 26.0.2 for Windows.
- Device model: PTM 65-nm bulk CMOS beta model released 22 February 2006.
- Nominal condition: 1.2 V, 27 C, 125 MHz, 10 fF per observed output.
- NMOS: L=65 nm, W=120 nm.
- PMOS: L=65 nm, W=240 nm.
- Input edge time: 20 ps.
- XOR transient maximum timestep: 2 ps.
- Encoder transient maximum timestep: 20 ps.
- Power: average of $-V(VDD) \cdot I(VSUP)$ over the documented steady-state window.
- Delay: 50%-to-50%; $tpd = (tpLH + tpHL) / 2$.
- Extended encoder delay endpoint: overall parity P0.

Sweeps

The experiment set uses independent one-factor sweeps around the nominal condition: VDD = 0.6, 0.8, 1.0, 1.2 V; temperature = -20, 27, 85 C; load = 1, 5, 10, 20 fF; frequency = 25, 125, 200 MHz. A redundant full Cartesian product is intentionally avoided.

Functional criteria

For the automated operating map, a measured point is considered functionally acceptable when V_{OH} is at least 0.8 VDD, V_{OL} is at most 0.2 VDD, and propagation delay is less than half the input period. This is an explicit study criterion, not a claim that it is a foundry-qualified noise-margin specification.

Traceability

Every generated netlist and log has an experiment ID. The manifest is ``data/experiments.json``; raw LTspice logs stay next to their circuits; parsed results are in ``data/processed/results.csv``; web data are in ``thesis/data/results.json``.

Nominal Comparison

Architecture	Power (uW)	Delay (ps)	PDP (fJ)	Swing (V)	Transistors	Status
CMOS	27.0237	344.947	9.3217	1.2064	264	Pass
FSGDI	2.5502	429.062	1.0942	1.2135	132	Pass
HYBRID_A	25.8367	278.783	7.2029	1.2039	220	Pass
HYBRID_B	11.2029	215.846	2.4181	1.2042	152	Pass

Hybrid-B reduces nominal power by 58.54%, delay by 37.43%, PDP by 74.06%, and transistor count by 42.42% relative to CMOS.

FS-GDI has the lowest nominal power (2.550 uW) and PDP (1.094 fJ), but its P0 path is slower than CMOS.

Robustness and Rejected Points

The one-factor sweeps cover 0.6-1.2 V, -20 to 85 C, 1-20 fF, and 25-200 MHz. Three 1 fF points are rejected by the automated supply-bound check because of overshoot. No failed point is replaced or hidden.